

Levels of measurement

- Statistics begins with defining and identifying the population of interest, which includes individuals, objects, or phenomena that the researcher intends to study.
- The researchers examine various characteristics of this population, referred to as **variables**.
- Once these variables are identified, statistical analysis is applied to describe the population and explore the relationships among the variables.
- When organizing and analyzing data, a critical step is to determine the scale used to measure the variables.
- Measurement scales can vary widely, from established units like kilograms for weight or centimeters for length, to subjective scales such as “strongly agree,” “agree,” “neutral,” “disagree,” and “strongly disagree” for opinions.
- In statistical analysis, variables are typically measured using one of four types of scales: **nominal, ordinal, interval, and ratio**.

Nominal scale

- A **nominal** scale is a scale which classifies the various values of the variable into categories. These categories are usually given names, or possibly numbers.
- Some examples of variables which are considered to be measured by nominal scales are **sex, ethnicity, religion, nationality, political party preferred and province of residence**.
- Sex can take on only two values, male and female.
- Ethnicity or nationality can be classified into categories like **English, French, Aboriginal, Ukrainian, and Vietnamese**, but again different ethnicities amount to no more than a classification of culture or nationality.
- The categories could alternatively be given numbers. For example, male could be given value 1 and female the value 2. These numbers may be used by data analysts when coding nominal scales for data entry to a computer. However, these codes are used only to keep track of the values male and female, and have no meaning as numerical values.
- Examples of nominal scale measurements
 - **Gender:** Male, Female, Non-binary
 - **Nationality:** American, Canadian, Indian, Australian
 - **Ethnicity:** Hispanic, Asian, African, European
 - **Religion:** Christianity, Islam, Hinduism, Buddhism
 - **Political party affiliation:** Democrat, Republican, Independent
 - **Favorite color:** Red, Blue, Green, Yellow
 - **Marital status:** Single, Married, Divorced, Widowed
 - **Blood group:** A, B, AB, O
 - **Mode of transport:** Car, Bus, Train, Bicycle

Ordinal scale

- An ordinal scale is a scale which is nominal, and one in which each value of the variable can be ordered, or ranked, as more than, less than, or equal to any other value of the variable.
- If a scale is ordinal, but no more than ordinal, then the different values of the scale can be ranked or ordered, but the distances between the values cannot be meaningfully determined.
- Examples of ordinal scale measurements
 - **Order of finish in a race:** First, Second, Third
 - **Customer satisfaction ratings:** Very Satisfied, Satisfied, Neutral, Dissatisfied, Very Dissatisfied
 - **Educational qualification:** High School, Bachelor’s Degree, Master’s Degree, Ph.D.

- **Socioeconomic status:** Low, Middle, High
- **Severity of pain:** No Pain, Mild, Moderate, Severe, Extreme
- **Grades in a course:** A, B, C, D, F
- **Star ratings for products or services:** 1-star, 2-stars, 3-stars, 4-stars, 5-stars
- **Performance ratings:** Excellent, Good, Average, Poor, Very Poor
- **Agreement levels in surveys:** Strongly Agree, Agree, Neutral, Disagree, Strongly Disagree
- **Weather severity levels:** Mild, Moderate, Severe, Extreme

Interval scale

- Consider that one person is taller than another person, the amount of difference in height can be measured. A person's height is certainly ordinal in nature, but with the additional characteristic that height differences can be meaningfully determined. The idea of a meaningful difference is the basis for the definition of an **interval level scale**.
- An **interval scale** of measurement is a scale in which equal numerical differences represent equal quantities or magnitudes of the characteristic.
- Example: The following table shows the heights and attitudes of four individuals:

Name	Height (cm)	Attitude
James	177	3 (Agree Somewhat)
John	180	2 (Disagree Somewhat)
Kyle	188	1 (Strongly Disagree)
Blaine	191	2 (Disagree Somewhat)

The height differences are consistent: John is 3 cm taller than James, and Blaine is 3 cm taller than Kyle. These equal differences (3 cm) have a constant, meaningful magnitude, making height an interval scale. On the other hand, the numerical difference between *Agree Somewhat* (3) and *Disagree Somewhat* (2) may not be the same as the difference between *Disagree Somewhat* (2) and *Strongly Disagree* (1). While the values can be ranked, the differences between them are not uniform or measurable, making attitudes an ordinal scale.

- An interval scale is alternatively defined as a scale which has a well defined unit of measure.
- One key characteristic of interval scale is it has no true zero value (e.g., zero does not imply “nothing”).
- Examples of interval scale measurements
 - **Temperature:** Measured in degrees Celsius or Fahrenheit (e.g., 0°C, 25°C, 100°C).
 - **Calendar Years:** Years such as 1990, 2000, 2025.
 - **Time of Day:** Measured in hours or minutes on a 24-hour clock (e.g., 10:00 AM, 3:30 PM).
 - **IQ Scores:** Intelligence scores (e.g., 90, 100, 120).
 - **SAT Scores:** Standardized test scores (e.g., 500, 600, 800).
 - **pH Levels:** Acidity or alkalinity measured on a scale from 0 to 14.
 - **Credit Scores:** Scores measuring creditworthiness (e.g., 650, 700, 800).
 - **Income Tax Brackets:** Measured in ranges such as \$10,000–\$20,000, \$20,000–\$30,000.
 - **Longitude/Latitude:** Geographic coordinates without a true zero (e.g., 30°N, 75°E).
 - **Exam Scores:** Graded on a percentage scale (e.g., 50%, 75%, 90%).

Ratio scale

- The final commonly used scale of measurement is a scale for which one can legitimately take the ratios of two values. This type of scale will at least be of the interval level, so that differences of values are meaningful, but with the additional characteristic that the ratio of two values will also be a meaningful magnitude.
- A **ratio** scale of measurement is an interval scale with the additional property that equal ratios between two possible values of a variable represent equal magnitudes.

- A **ratio scale** allows meaningful comparisons of ratios between values, as it includes a true zero point. For example:
 - A monthly income of \$3,000 is twice that of \$1,500.
 - Similarly, an income of \$1,000 is twice that of \$500.

In each case, the ratio (2:1) holds true, making the scale consistent for comparisons.

- While most interval scales can also be ratio scales, some lack a true zero. For instance:
 - **Temperature in Celsius or Fahrenheit:** These scales have arbitrary zero points (e.g., 0°C is the freezing point of water, not the absence of temperature). Ratios on these scales are not meaningful; for example, 40°C is not "twice as hot" as 20°C.

In contrast, ratio scales, such as income or weight, allow for both meaningful differences and ratios due to their absolute zero.

- Examples of ratio scale measurements
 - **Income:** Monthly or yearly income (e.g., \$1,000, \$2,000, \$3,000).
 - **Weight:** Measured in kilograms or pounds (e.g., 50 kg, 100 kg).
 - **Height:** Measured in centimeters or inches (e.g., 150 cm, 180 cm).
 - **Distance:** Measured in meters, kilometers, or miles (e.g., 5 km, 10 km).
 - **Time:** Measured in seconds, minutes, or hours (e.g., 10 minutes, 20 minutes).
 - **Age:** Measured in years (e.g., 10 years, 25 years).
 - **Speed:** Measured in km/h or mph (e.g., 60 km/h, 120 km/h).
 - **Number of Students:** Total students in a class (e.g., 20, 30, 50).
 - **Energy Consumption:** Measured in kilowatt-hours (e.g., 100 kWh, 200 kWh).
 - **Population:** Number of people in a city or country (e.g., 1,000,000, 2,000,000).

Examples

1. Last weekend, Sarah decided to take a break and travel to a nearby hill station. She visited three locations: the waterfall, the botanical garden, and the sunset point, in that order of preference. At each spot, she recorded the time she spent, such as 2 hours at the waterfall, 1 hour at the botanical garden, and 30 minutes at the sunset point. She also noted the ticket prices: \$10 for the botanical garden and \$5 for the sunset point, while the waterfall was free. To make her trip more efficient, she ranked the spots based on accessibility, with the waterfall being the easiest to reach. Identify the variables in this story and the scale of measurement for each variable.
2. David started a fitness program to improve his health. He weighed 80 kg at the beginning and aimed to lose 10 kg in three months. He attended three types of fitness classes: yoga, weightlifting, and swimming, which he ranked as swimming > weightlifting > yoga in terms of his preference. He tracked his progress weekly by noting the number of hours spent exercising (e.g., 6 hours in the first week, 8 hours in the second week) and recording his weight changes. David also marked his gym visits in a journal and labeled them as Morning or Evening sessions. List the variables and determine the scale of measurement for each variable.
3. Maria visited a hospital for her annual check-up. The hospital recorded her personal details, including her blood type (A, B, AB, or O) and her age in years (32). During the visit, the doctor measured her height (165 cm), weight (60 kg), and blood pressure (120/80 mmHg). Maria also rated her experience at the hospital on a scale from 1 (Very Poor) to 5 (Excellent). The hospital noted the department she visited (General Medicine) and the time she spent waiting, which was 45 minutes. Identify all the variables recorded during Maria's hospital visit. For each variable, determine its scale of measurement (nominal, ordinal, interval, or ratio). Explain why the scale of measurement is appropriate for each variable.